PM Accelerator Technical Assessment

Weather Trend Forecasting

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Our Mission

By making industry-leading tools and education available to individuals from all backgrounds, we level the playing field for future PM leaders. This is the PM Accelerator motto, as we grant aspiring and experienced PMs what they need most – Access. We introduce you to industry leaders, surround you with the right PM ecosystem, and discover the new world of AI product management skills.

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1. Imports & Setup

In [1]:
```
import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import geopandas as gpd
import shap
import kagglehub
import tensorflow as tf

from scipy import stats
from sklearn.preprocessing import MinMaxScaler
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense
```

2. Data Loading

In [2]:
```
path = kagglehub.dataset_download("nelgiriyewithana/global-weather-repository")
print("Dataset path:", path)
print("Files:", os.listdir(path))
```

```
Downloading to /root/.cache/kagglehub/datasets/nelgiriyewithana/global-weather-repos
itory/936.archive...
100%|██████████| 10.9M/10.9M [00:00<00:00, 12.9MB/s]
Extracting files...
Dataset path: /root/.cache/kagglehub/datasets/nelgiriyewithana/global-weather-reposi
tory/versions/936
Files: ['state.db', 'GlobalWeatherRepository.csv']
```

In [40]:
```
path_data = os.path.join(path, "GlobalWeatherRepository.csv")
df = pd.read_csv(path_data)
print(f"Shape: {df.shape}")
df.head()
```

```
Shape: (137803, 41)
```

Out[40]:

	country	location_name	latitude	longitude	timezone	last_updated_epoch	las
0	Afghanistan	Kabul	34.52	69.18	Asia/Kabul	1715849100	2
1	Albania	Tirana	41.33	19.82	Europe/Tirane	1715849100	2
2	Algeria	Algiers	36.76	3.05	Africa/Algiers	1715849100	2
3	Andorra	Andorra La Vella	42.50	1.52	Europe/Andorra	1715849100	2
4	Angola	Luanda	-8.84	13.23	Africa/Luanda	1715849100	2

5 rows × 41 columns

3. Exploratory Data Analysis & Cleaning

3.1 Data Quality

In [41]:
```
print("=== dtypes ===")
print(df.dtypes)
```

```
=== dtypes ===
country                          object
location_name                    object
latitude                        float64
longitude                       float64
timezone                         object
last_updated_epoch                int64
last_updated                     object
temperature_celsius             float64
temperature_fahrenheit          float64
condition_text                   object
wind_mph                        float64
wind_kph                        float64
wind_degree                       int64
wind_direction                   object
pressure_mb                     float64
pressure_in                     float64
precip_mm                       float64
precip_in                       float64
humidity                          int64
cloud                             int64
feels_like_celsius              float64
feels_like_fahrenheit           float64
visibility_km                   float64
visibility_miles                float64
uv_index                        float64
gust_mph                        float64
gust_kph                        float64
air_quality_Carbon_Monoxide     float64
air_quality_Ozone               float64
air_quality_Nitrogen_dioxide    float64
air_quality_Sulphur_dioxide     float64
air_quality_PM2.5               float64
air_quality_PM10                float64
air_quality_us-epa-index          int64
air_quality_gb-defra-index        int64
sunrise                          object
sunset                           object
moonrise                         object
moonset                          object
moon_phase                       object
moon_illumination                 int64
dtype: object
```

In [42]:

```python
print("=== Missing values ===")
df.isna().sum()
```

```
=== Missing values ===
```

Out[42]:

	0
country	0
location_name	0
latitude	0
longitude	0
timezone	0
last_updated_epoch	0
last_updated	0
temperature_celsius	0
temperature_fahrenheit	0
condition_text	0
wind_mph	0
wind_kph	0
wind_degree	0
wind_direction	0
pressure_mb	0
pressure_in	0
precip_mm	0
precip_in	0
humidity	0
cloud	0
feels_like_celsius	0
feels_like_fahrenheit	0
visibility_km	0
visibility_miles	0
uv_index	0
gust_mph	0
gust_kph	0
air_quality_Carbon_Monoxide	0
air_quality_Ozone	0
air_quality_Nitrogen_dioxide	0

	0
air_quality_Sulphur_dioxide	0
air_quality_PM2.5	0
air_quality_PM10	0
air_quality_us-epa-index	0
air_quality_gb-defra-index	0
sunrise	0
sunset	0
moonrise	0
moonset	0
moon_phase	0
moon_illumination	0

dtype: int64

In [43]:
```
print("=== Duplicated rows ===")
df.duplicated().sum()
```

```
=== Duplicated rows ===
```

Out[43]:
```
np.int64(0)
```

- No missing values and no duplicate rows across all features.

3.2 Object Column Audit

Identify which object columns encode numeric or datetime values so they can be cast correctly.

In [5]:
```
obj_cols = df.select_dtypes(include=['object']).columns.tolist()
summary = []
for c in obj_cols:
    s = df[c].astype(str).str.strip()
    numeric_try = pd.to_numeric(s, errors='coerce')
    datetime_try = pd.to_datetime(s, errors='coerce', utc=False)
    summary.append({
        'column': c,
        'unique': df[c].nunique(dropna=True),
        'numeric_parse_rate': round(numeric_try.notna().mean(), 3),
        'datetime_parse_rate': round(datetime_try.notna().mean(), 3),
        'sample_values': ', '.join(df[c].dropna().astype(str).head(3).tolist())
    })
pd.DataFrame(summary).sort_values(['numeric_parse_rate', 'datetime_parse_rate'], as
```

```
/tmp/ipykernel_5430/1741366338.py:6: UserWarning: Could not infer format, so each el
ement will be parsed individually, falling back to `dateutil`. To ensure parsing is
consistent and as-expected, please specify a format.
  datetime_try = pd.to_datetime(s, errors='coerce', utc=False)
/tmp/ipykernel_5430/1741366338.py:6: UserWarning: Could not infer format, so each el
ement will be parsed individually, falling back to `dateutil`. To ensure parsing is
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/tmp/ipykernel_5430/1741366338.py:6: UserWarning: Could not infer format, so each el
ement will be parsed individually, falling back to `dateutil`. To ensure parsing is
consistent and as-expected, please specify a format.
  datetime_try = pd.to_datetime(s, errors='coerce', utc=False)
```

Out[5]:

	column	unique	numeric_parse_rate	datetime_parse_rate	sample_values
3	last_updated	22925	0.0	1.000	2024-05-16 13:15, 2024-05-16 10:45, 2024-05-16...
6	sunrise	495	0.0	1.000	04:50 AM, 05:21 AM, 05:40 AM
7	sunset	547	0.0	1.000	06:50 PM, 07:54 PM, 07:50 PM
8	moonrise	1444	0.0	0.966	12:12 PM, 12:58 PM, 01:15 PM
9	moonset	1444	0.0	0.966	01:11 AM, 02:14 AM, 02:14 AM
0	country	211	0.0	0.000	Afghanistan, Albania, Algeria
1	location_name	257	0.0	0.000	Kabul, Tirana, Algiers
2	timezone	199	0.0	0.000	Asia/Kabul, Europe/Tirane, Africa/Algiers
4	condition_text	49	0.0	0.000	Partly Cloudy, Partly cloudy, Sunny
5	wind_direction	16	0.0	0.000	NNW, NW, W
10	moon_phase	8	0.0	0.000	Waxing Gibbous, Waxing Gibbous, Waxing Gibbous

Observations:

- `last_updated`, `sunrise`, and `sunset` parse cleanly as datetime (100% parse rate) and will be converted to structured time features.
- `moonrise` and `moonset` parse at ~96.6% — the remainder are likely "No moonrise/moonset" strings that will be filled with 0.
- `country`, `location_name`, `timezone`, `condition_text`, `wind_direction`, and `moon_phase` are true categoricals.

3.3 Data Cleaning

- Drop the redundant epoch column (already have `last_updated` as a string).
- Standardise country names that appear in multiple languages/transliterations.

In [6]:
```python
# Drop redundant epoch column
df = df.drop('last_updated_epoch', axis=1)
```

In [7]:
```python
# Identify rare country spellings (< 10 rows) before normalising
country_counts = df['country'].value_counts(dropna=False)
print("Rare country spellings:")
print(country_counts[country_counts < 10])
```

```
Rare country spellings:
country
USA United States of America        3
Lao People's Democratic Republic    1
Libya                               1
Kosovo                              1
Togo                                1
Malásia                             1
Komoren                             1
1                             كولومبيا
Colombia                            1
Estonie                             1
Гватемала                           1
Letonia                             1
Inde                                1
Mexique                             1
Polônia                             1
Marrocos                            1
Польша                              1
Турция                              1
Saint-Vincent-et-les-Grenadines     1
Südkorea                            1
Saudi Arabien                       1
Bélgica                             1
Turkménistan                        1
火鸡                                   1
Jemen                               1
Name: count, dtype: int64
```

- Several country names appear in non-English scripts (Arabic, Russian, Portuguese, German, French)

In [8]:
```python
# Normalize country names
translation = {
    'Malásia': 'Malaysia',
    'كولومبيا': 'Colombia',
    'Гватемала': 'Guatemala',
    'Польша': 'Poland',
    'Polônia': 'Poland',
    'Турция': 'Turkey',
    'Südkorea': 'South Korea',
    'Bélgica': 'Belgium',
    'Turkménistan': 'Turkmenistan',
    '火鸡': 'Turkey',
    'USA United States of America': 'United States of America',
    'Komoren': 'Comoros',
```

```
    'Estonie': 'Estonia',
    'Inde': 'India',
    'Mexique': 'Mexico',
    'Letonia': 'Latvia',
    'Marrocos': 'Morocco',
    'Saudi Arabien': 'Saudi Arabia',
    'Jemen': 'Yemen',
    'Saint-Vincent-et-les-Grenadines': 'Saint Vincent and the Grenadines',
    'Kyrghyzstan': 'Kyrgyzstan',
    "Cote d'Ivoire": "Côte d'Ivoire"
}
df['country'] = df['country'].replace(translation)
print('Unique countries:', df['country'].nunique())
```

```
Unique countries: 191
```

3.4 Numeric Summary & Outlier Detection

For each numeric column, compute descriptive statistics and flag outliers via both the IQR rule and the Z-score rule (|z| > 3).

In [9]:

```
# Calculating important measures for each numeric column
numeric_df = df.select_dtypes(include=[np.number]).copy()

# Per-column quartiles and IQR
q1 = numeric_df.quantile(0.25)
q3 = numeric_df.quantile(0.75)
iqr = q3 - q1
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr

# IQR outlier mask/counts
iqr_outlier_mask = (numeric_df.lt(lower, axis=1)) | (numeric_df.gt(upper, axis=1))
iqr_outlier_counts = iqr_outlier_mask.sum()

# Z-score outlier mask/counts (abs(z) > 3)
z_scores = pd.DataFrame(
    stats.zscore(numeric_df, nan_policy='omit'),
    columns=numeric_df.columns,
    index=numeric_df.index
).abs()
z_outlier_mask = z_scores > 3
z_outlier_counts = z_outlier_mask.sum()

# Summary table like your original stats_df
stats_df = pd.DataFrame({
    'Mean': numeric_df.mean(),
    'Median': numeric_df.median(),
    'Std': numeric_df.std(),
    'Min': numeric_df.min(),
    'Max': numeric_df.max(),
    'Range': numeric_df.max() - numeric_df.min(),
    'IQR': iqr,
    'Outliers-IQR': iqr_outlier_counts,
    'Outliers-ZScore': z_outlier_counts
```

```
}).reset_index(names='Column')

# Optional percentages for quick prioritization
stats_df['Outliers-IQR-%'] = (100 * stats_df['Outliers-IQR'] / len(df)).round(2)
stats_df['Outliers-ZScore-%'] = (100 * stats_df['Outliers-ZScore'] / len(df)).round

# Sort by most problematic columns
stats_df = stats_df.sort_values(['Outliers-IQR', 'Outliers-ZScore'], ascending=Fals

# If you need actual rows later (not printed), use these dicts:
outlier_rows_iqr = {
    c: df.loc[iqr_outlier_mask[c], [c]].copy()
    for c in numeric_df.columns
}
outlier_rows_zscore = {
    c: df.loc[z_outlier_mask[c], [c]].copy()
    for c in numeric_df.columns
}

stats_df
```

Out[9]:

	Column	Mean	Median	Std	Min	Max
15	visibility_km	9.507689	10.0000	2.692913	0.000	32.000
16	visibility_miles	5.616242	6.0000	1.682258	0.000	19.000
9	precip_mm	0.133865	0.0000	0.560860	0.000	42.240
10	precip_in	0.005073	0.0000	0.022168	0.000	1.660
23	air_quality_Sulphur_dioxide	10.278280	2.4050	35.368065	-9999.000	521.330
25	air_quality_PM10	48.166512	19.8500	148.501486	-1848.150	6037.290
22	air_quality_Nitrogen_dioxide	14.942860	5.6500	23.587785	0.000	427.700
27	air_quality_gb-defra-index	2.596518	2.0000	2.436361	1.000	10.000
20	air_quality_Carbon_Monoxide	457.461781	292.3000	753.322375	-9999.000	38879.398
24	air_quality_PM2.5	24.088876	14.0600	36.652515	0.168	1614.100
1	longitude	21.953131	23.2361	65.786751	-175.200	179.220
26	air_quality_us-epa-index	1.695442	1.0000	0.937895	1.000	6.000
8	pressure_in	29.944578	29.9300	0.304043	27.960	88.770
7	pressure_mb	1014.059839	1013.0000	10.297742	947.000	3006.000
18	gust_mph	11.325028	9.5000	8.554724	2.200	1845.700
19	gust_kph	18.227747	15.3000	13.767419	3.600	2970.400
21	air_quality_Ozone	57.920902	55.0000	30.659883	0.000	480.700
13	feels_like_celsius	22.107862	25.1000	11.584761	-36.700	81.300
14	feels_like_fahrenheit	71.788717	77.2000	20.850200	-34.000	178.300
2	temperature_celsius	21.256764	23.9000	9.681062	-29.800	79.300
5	wind_kph	12.886087	10.8000	11.559789	3.600	2963.200
3	temperature_fahrenheit	70.263959	75.0000	17.425782	-21.600	174.700
4	wind_mph	8.004772	6.7000	7.184921	2.200	1841.200
17	uv_index	3.303729	1.9000	3.546056	0.000	16.300
0	latitude	19.212246	17.2500	24.414995	-41.300	64.150
6	wind_degree	168.800781	161.0000	103.790368	1.000	360.000
11	humidity	66.685246	72.0000	23.857822	2.000	100.000
12	cloud	39.940836	30.0000	34.091114	0.000	100.000
28	moon_illumination	49.448016	49.0000	35.012652	0.000	100.000

Observations:

- `visibility_km` and `visibility_miles` have the highest IQR outlier count (~21-22% of rows) but this is expected, visibility varies enormously by weather event and is not a data error.
- `air_quality_Carbon_Monoxide` and `air_quality_Sulphur_dioxide` contain values of **-9999**, a sentinel for missing measurements. These should be treated as missing rather than true extremes.
- `precip_mm` and `precip_in` are heavily right-skewed (median = 0, max in the hundreds), precipitation is inherently sparse and bursty, so high IQR outlier counts here are expected.
- `gust_mph` and `gust_kph` show physically impossible maxima (~1845 mph / ~2970 kph). These are sensor errors and warrant capping or removal before modelling.
- `latitude`, `wind_degree`, `humidity`, `cloud`, and `moon_illumination` show **zero outliers**, they are bounded variables by definition.
- Z-score outliers are rare across the board (<4%), which suggests the distributions, while skewed, don't have extreme isolated spikes beyond the sensor errors noted above.

4. Feature Engineering

- Parse `last_updated` → calendar features (year, month, day, weekday, hour)
- Cyclically encode `hour` and `month` to preserve their periodic structure
- MinMax-scale all numeric features so they share the same [0, 1] range for the neural network
- Parse sunrise/sunset/moonrise/moonset strings to hour-of-day integers

In [10]:
```
scaler = MinMaxScaler()

# Parse datetime and extract calendar features
df['last_updated'] = pd.to_datetime(df['last_updated'])
df['year']    = df['last_updated'].dt.year
df['month']   = df['last_updated'].dt.month
df['day']     = df['last_updated'].dt.day
df['weekday'] = df['last_updated'].dt.day_name()
df['hour']    = df['last_updated'].dt.hour

# Cyclical encoding keeps hour 23 → hour 0 distance small
df['Hour_sin']  = np.sin(2 * np.pi * df['hour']  / 24)
df['Hour_cos']  = np.cos(2 * np.pi * df['hour']  / 24)
df['Month_sin'] = np.sin(2 * np.pi * df['month'] / 12)
df['Month_cos'] = np.cos(2 * np.pi * df['month'] / 12)

# Columns to scale: all original numeric + the 4 cyclical encodings
numeric_df = df.select_dtypes(include=[np.number]).copy()
columns = numeric_df.columns.tolist() + ['Hour_sin', 'Hour_cos', 'Month_sin', 'Mont
# deduplicate preserving order (Hour_sin etc. already added to df)
```

```
columns = list(dict.fromkeys(columns))

df_numeric_normalized = pd.DataFrame(
    scaler.fit_transform(df[columns]), columns=columns
)

# Combine with categorical columns
df_scaled = df.select_dtypes(exclude=['number']).copy()
df_scaled[columns] = df_numeric_normalized.values

print("Scaled dataframe shape:", df_scaled.shape)
df_scaled.head()
```

```
Scaled dataframe shape: (137803, 49)
```

Out[10]:

	country	location_name	timezone	last_updated	condition_text	wind_direction
0	Afghanistan	Kabul	Asia/Kabul	2024-05-16 13:15:00	Partly Cloudy	NNW
1	Albania	Tirana	Europe/Tirane	2024-05-16 10:45:00	Partly cloudy	NW
2	Algeria	Algiers	Africa/Algiers	2024-05-16 09:45:00	Sunny	W
3	Andorra	Andorra La Vella	Europe/Andorra	2024-05-16 10:45:00	Light drizzle	SW
4	Angola	Luanda	Africa/Luanda	2024-05-16 09:45:00	Partly cloudy	SSE

5 rows × 49 columns

In [11]:

```
def parse_time_to_hour(series):
    """Convert strings like '04:50 AM' to hour-of-day integers.
    Missing / unparseable values fall back to 0.
    """
    parsed = pd.to_datetime(series, format='%I:%M %p', errors='coerce')
    return parsed.dt.hour.fillna(0).astype(int)

for col in ['sunrise', 'sunset', 'moonrise', 'moonset']:
    df_scaled[col] = parse_time_to_hour(df[col])

df_scaled.head()
```

Out[11]:

	country	location_name	timezone	last_updated	condition_text	wind_direction
0	Afghanistan	Kabul	Asia/Kabul	2024-05-16 13:15:00	Partly Cloudy	NNW
1	Albania	Tirana	Europe/Tirane	2024-05-16 10:45:00	Partly cloudy	NW
2	Algeria	Algiers	Africa/Algiers	2024-05-16 09:45:00	Sunny	W
3	Andorra	Andorra La Vella	Europe/Andorra	2024-05-16 10:45:00	Light drizzle	SW
4	Angola	Luanda	Africa/Luanda	2024-05-16 09:45:00	Partly cloudy	SSE

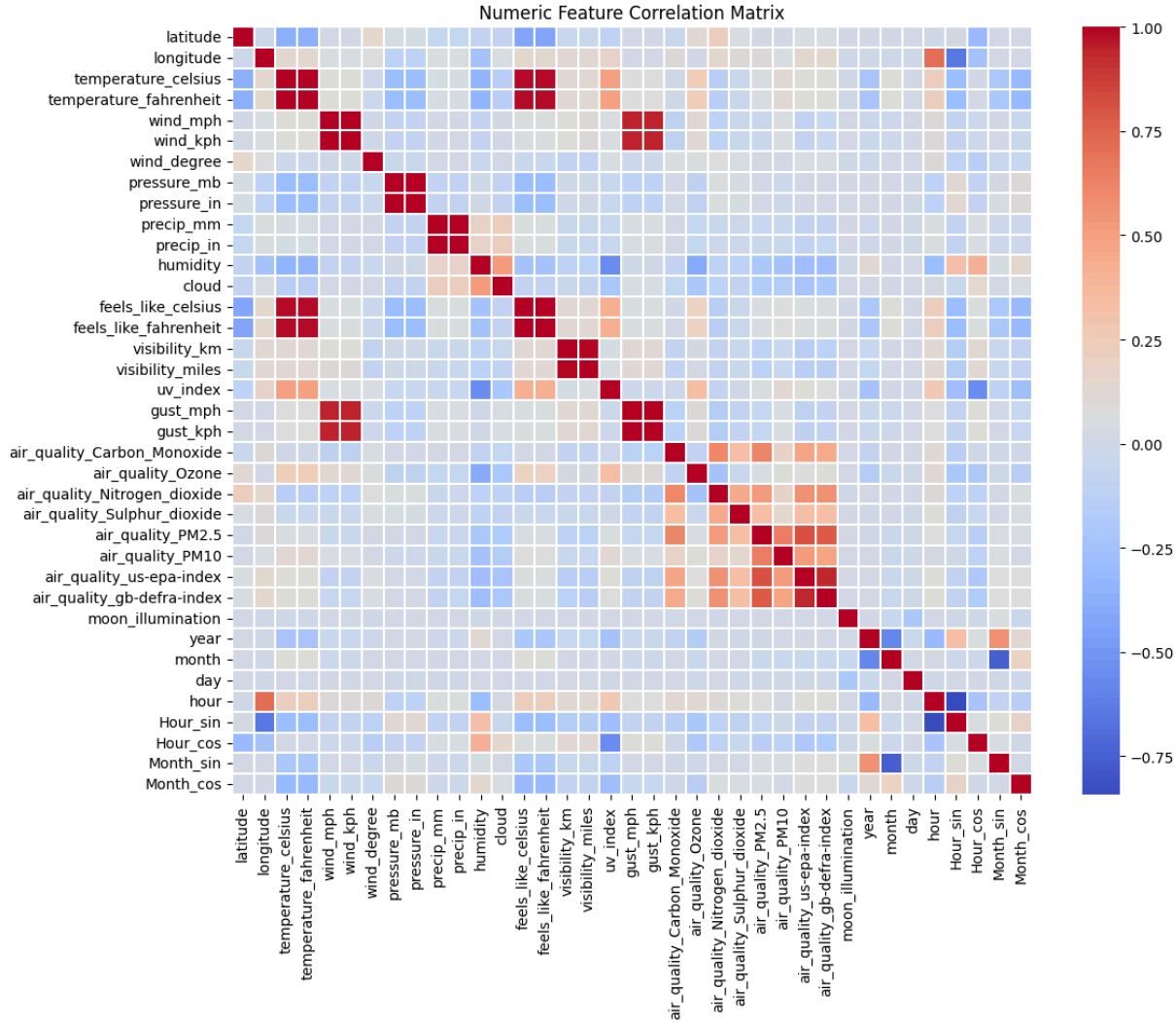
5 rows × 49 columns

5. Exploratory Visualisations

5.1 Correlation Matrix

In [12]:

```
correlation_matrix = df_scaled[columns].corr()
plt.figure(figsize=(12, 10))
sns.heatmap(correlation_matrix, annot=False, cmap='coolwarm', linewidths=0.3)
plt.title("Numeric Feature Correlation Matrix")
plt.tight_layout()
plt.show()
```

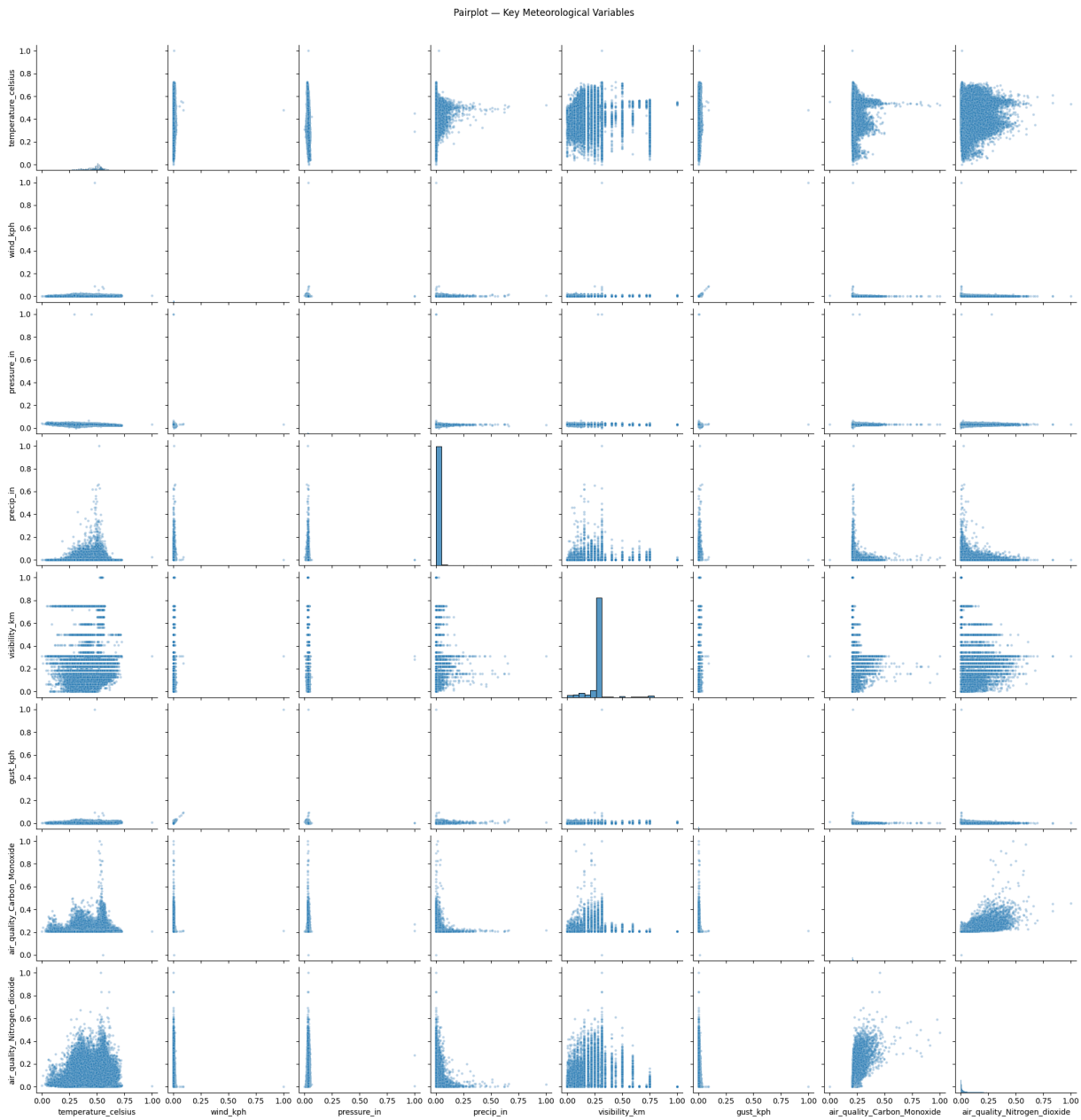


Observations:

- The strongest correlations are between unit-duplicate pairs: `temperature_celsius` / `temperature_fahrenheit`, `wind_kph` / `wind_mph`, `pressure_mb` / `pressure_in`, `visibility_km` / `visibility_miles`, `gust_kph` / `gust_mph`, `feels_like_celsius` / `feels_like_fahrenheit`. These are perfectly collinear by construction — in a production model, one from each pair should be dropped.
- The air quality cluster (`CO`, `NO₂`, `SO₂`, `PM2.5`, `PM10`, `us-epa-index`, `gb-defra-index`) shows moderate internal correlation, suggesting a shared pollution signal.
- `Hour_sin` / `Hour_cos` and `Month_sin` / `Month_cos` show expected anti-correlation with each other this is the correct behaviour of cyclical encoding.
- Temperature features correlate positively with `uv_index` and negatively with `humidity`, matching physical intuition.
- Most air quality features are weakly correlated with meteorological variables, suggesting they carry independent signal.

5.2 Pairplot — Key Meteorological Variables

In [13]:
```
key_vars = ['temperature_celsius', 'wind_kph', 'pressure_in',
            'precip_in', 'visibility_km', 'gust_kph',
            'air_quality_Carbon_Monoxide', 'air_quality_Nitrogen_dioxide']
sns.pairplot(df_scaled[key_vars], plot_kws={'alpha': 0.3, 's': 10})
plt.suptitle("Pairplot — Key Meteorological Variables", y=1.02)
plt.show()
```



Observations:

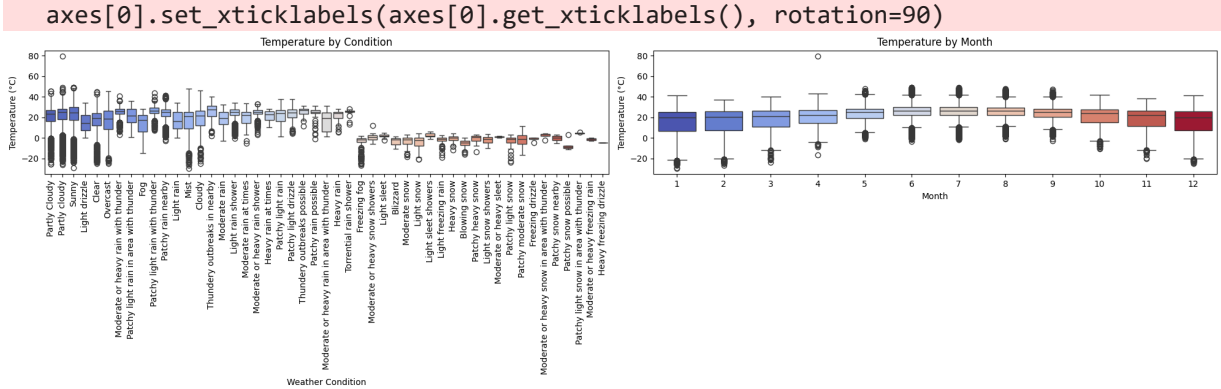
- The pairplot confirms heavy right-skew in `precip_in`, the vast majority of observations cluster near zero, with a long tail of precipitation events.
- `pressure_in` shows a bimodal-ish distribution, likely reflecting the mix of high-altitude and sea-level stations in a global dataset.
- `wind_kph` and `gust_kph` are strongly correlated and similarly skewed, with a handful of extreme values.

- No clear linear relationships emerge between air quality variables and meteorological ones, consistent with the correlation matrix.

5.3 Temperature Distribution

In [14]:
```
fig, axes = plt.subplots(1, 2, figsize=(20, 6))

sns.boxplot(x=df['condition_text'], y=df['temperature_celsius'],
            hue=df['condition_text'], palette='coolwarm', legend=False, ax=axes[0])
axes[0].set_xticklabels(axes[0].get_xticklabels(), rotation=90)
axes[0].set_xlabel("Weather Condition")
axes[0].set_ylabel("Temperature (°C)")
axes[0].set_title("Temperature by Condition")

sns.boxplot(x=df['month'], y=df['temperature_celsius'],
            hue=df['month'], palette='coolwarm', legend=False, ax=axes[1])
axes[1].set_xlabel("Month")
axes[1].set_ylabel("Temperature (°C)")
axes[1].set_title("Temperature by Month")

plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_5430/2477776188.py:5: UserWarning: set_ticklabels() should only be us
ed with a fixed number of ticks, i.e. after set_ticks() or using a FixedLocator.
  axes[0].set_xticklabels(axes[0].get_xticklabels(), rotation=90)
```

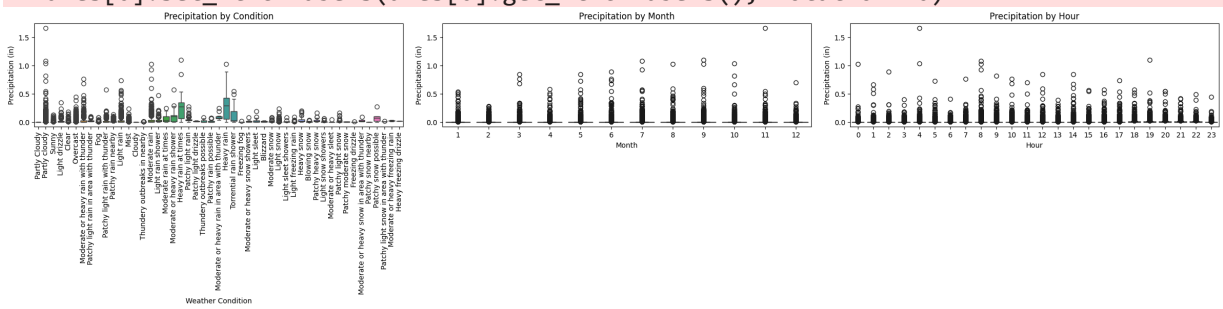


5.4 Precipitation Distribution

In [15]:
```
fig, axes = plt.subplots(1, 3, figsize=(24, 6))

sns.boxplot(x=df['condition_text'], y=df['precip_in'],
            hue=df['condition_text'], legend=False, ax=axes[0])
axes[0].set_xticklabels(axes[0].get_xticklabels(), rotation=90)
axes[0].set_xlabel("Weather Condition"); axes[0].set_ylabel("Precipitation (in)")
axes[0].set_title("Precipitation by Condition")

sns.boxplot(x=df['month'], y=df['precip_in'],
            hue=df['month'], legend=False, ax=axes[1])
axes[1].set_xlabel("Month"); axes[1].set_ylabel("Precipitation (in)")
axes[1].set_title("Precipitation by Month")
```

```
sns.boxplot(x=df['hour'], y=df['precip_in'],
            hue=df['hour'], legend=False, ax=axes[2])
axes[2].set_xlabel("Hour"); axes[2].set_ylabel("Precipitation (in)")
axes[2].set_title("Precipitation by Hour")

plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_5430/2500533363.py:5: UserWarning: set_ticklabels() should only be us
ed with a fixed number of ticks, i.e. after set_ticks() or using a FixedLocator.
  axes[0].set_xticklabels(axes[0].get_xticklabels(), rotation=90)
```

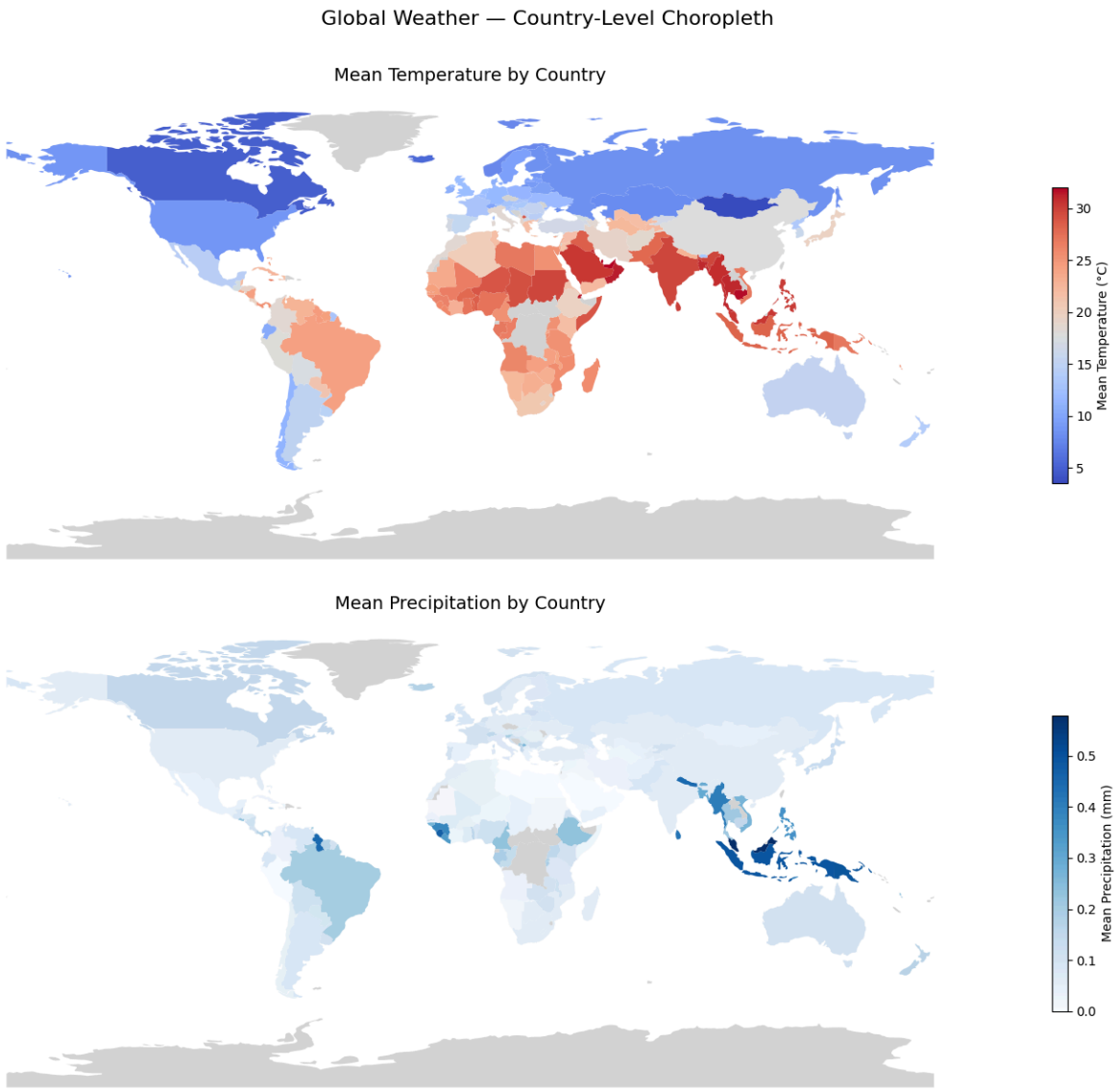


5.5 Geospatial Overview

In [16]:
```
SHAPEFILE = '/usr/local/lib/python3.12/dist-packages/pyogrio/tests/fixtures/natural
world = gpd.read_file(SHAPEFILE)

country_agg = (
    df.groupby('country', as_index=False)
      .agg(mean_temp=('temperature_celsius', 'mean'),
           mean_precip_mm=('precip_mm', 'mean'))
)

world_temp   = world.merge(country_agg[['country', 'mean_temp']],
                           left_on='name', right_on='country', how='left')
world_precip = world.merge(country_agg[['country', 'mean_precip_mm']],
                           left_on='name', right_on='country', how='left')

print(f"Country match coverage: {country_agg['country'].isin(world['name']).mean():

fig, axes = plt.subplots(2, 1, figsize=(16, 12))

world_temp.plot(column='mean_temp', ax=axes[0], cmap='coolwarm', legend=True,
                missing_kwds={'color': 'lightgrey', 'label': 'No data'},
                legend_kwds={'label': 'Mean Temperature (°C)', 'shrink': 0.6})
axes[0].set_title('Mean Temperature by Country', fontsize=14)
axes[0].axis('off')

world_precip.plot(column='mean_precip_mm', ax=axes[1], cmap='Blues', legend=True,
                  missing_kwds={'color': 'lightgrey', 'label': 'No data'},
                  legend_kwds={'label': 'Mean Precipitation (mm)', 'shrink': 0.6})
axes[1].set_title('Mean Precipitation by Country', fontsize=14)
axes[1].axis('off')

plt.suptitle('Global Weather — Country-Level Choropleth', fontsize=16, y=1.01)
```

```
plt.tight_layout()
plt.show()
```

```
Country match coverage: 80.1%
```



6. Location Focus: Tunis

Filter to a single station and plot the raw time series for the two targets we will forecast: temperature and precipitation.

Why a single location?

Training a time-series model on the global dataset without grouping by location would mix weather signals from different climates, geographies, and hemispheres, a sequence from Kabul followed by one from São Paulo has no temporal meaning. A location-level model learns a coherent climate signal.

Tunis is chosen as the focus station: it sits in a Mediterranean climate with clear seasonal temperature cycles and irregular precipitation, making it a good test case for both targets. The same pipeline can be applied to any other location in the dataset by changing the `location` variable.

In [17]:
```
location = 'Tunis'
df_loc = df_scaled[df_scaled['location_name'] == location].sort_values('last_update
print(f"{location}: {len(df_loc)} observations")
df_loc.head()
```

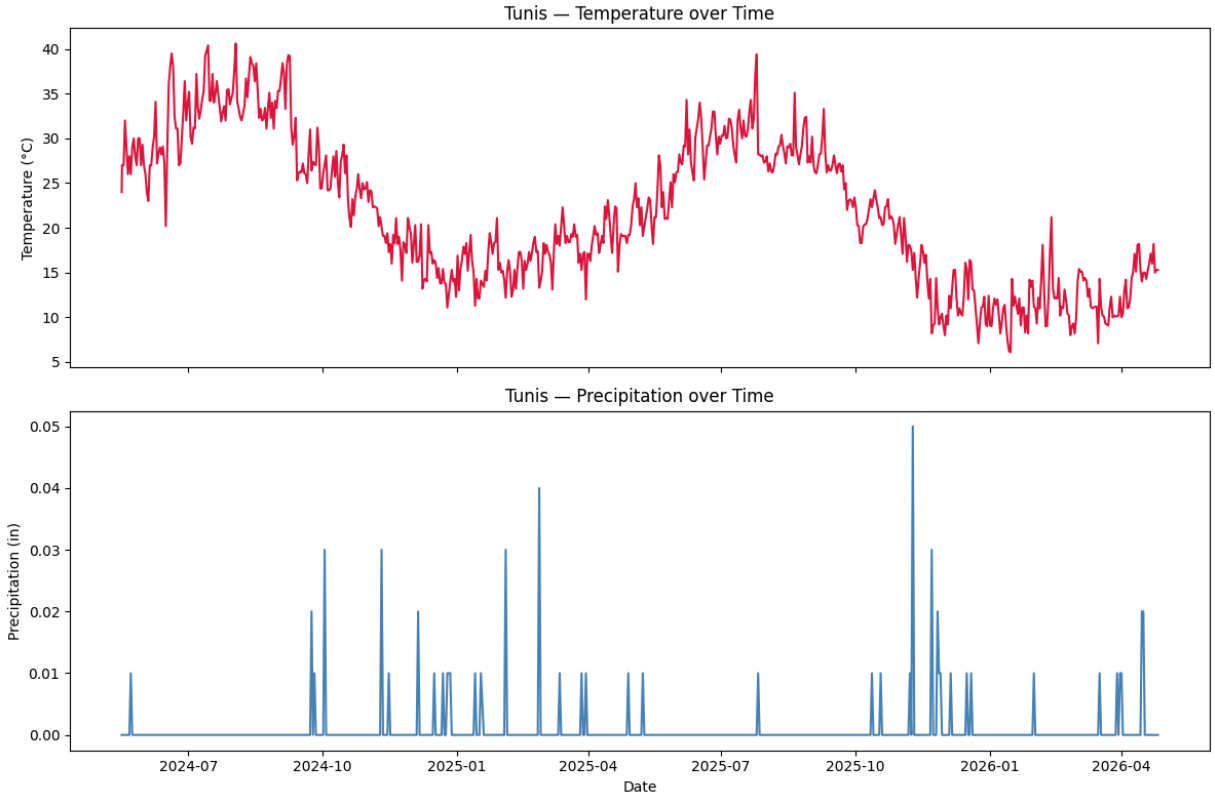
```
Tunis: 706 observations
```

Out[17]:

	country	location_name	timezone	last_updated	condition_text	wind_direction	sur
178	Tunisia	Tunis	Africa/Tunis	2024-05-16 09:45:00	Partly cloudy	WNW	
373	Tunisia	Tunis	Africa/Tunis	2024-05-16 15:15:00	Partly cloudy	NNW	
566	Tunisia	Tunis	Africa/Tunis	2024-05-17 17:00:00	Partly cloudy	ENE	
759	Tunisia	Tunis	Africa/Tunis	2024-05-18 15:30:00	Partly cloudy	E	
954	Tunisia	Tunis	Africa/Tunis	2024-05-19 15:15:00	Partly cloudy	ESE	

5 rows × 49 columns

In [18]:
```
temp_idx   = columns.index('temperature_celsius')
precip_idx = columns.index('precip_in')

def rescale(arr):
    """Inverse-transform a 2-D array back to original units."""
    return scaler.inverse_transform(arr)

fig, axes = plt.subplots(2, 1, figsize=(12, 8), sharex=True)

raw_values = rescale(df_loc[columns].values)
axes[0].plot(df_loc['last_updated'].values, raw_values[:, temp_idx], color='crimson
axes[0].set_ylabel("Temperature (°C)")
axes[0].set_title(f"{location} — Temperature over Time")

axes[1].plot(df_loc['last_updated'].values, raw_values[:, precip_idx], color='steel
axes[1].set_ylabel("Precipitation (in)")
axes[1].set_title(f"{location} — Precipitation over Time")
axes[1].set_xlabel("Date")

plt.tight_layout()
plt.show()
```



7. Sequence Creation & Train / Test Split

We use a sliding window of 5 time steps to predict the next step across all scaled features. A single chronological 80/20 split is applied

In [19]:
```
def create_sequences(data, time_steps=5):
    X, y = [], []
    for i in range(len(data) - time_steps):
        X.append(data[i : i + time_steps])
        y.append(data[i + time_steps])
    return np.array(X), np.array(y)

X, y = create_sequences(df_loc[columns].to_numpy())
print(f"Sequences: X={X.shape}, y={y.shape}")

# Canonical chronological split — shared by LSTM, RF and stacking
split_idx = int(len(X) * 0.8)

X_train_lstm, X_test_lstm = X[:split_idx], X[split_idx:]
y_train,       y_test      = y[:split_idx], y[split_idx:]

# Flattened inputs for tree-based model
X_flat = X.reshape(X.shape[0], X.shape[1] * X.shape[2])
X_train_rf, X_test_rf = X_flat[:split_idx], X_flat[split_idx:]

print(f"Train: {X_train_lstm.shape} | Test: {X_test_lstm.shape}")
```

```
Sequences: X=(701, 5, 37), y=(701, 37)
Train: (560, 5, 37) | Test: (141, 5, 37)
```

8. LSTM Model

In [20]:
```
model = Sequential([
    LSTM(units=248, return_sequences=False,
         input_shape=(X_train_lstm.shape[1], X_train_lstm.shape[2])),
    Dense(128),
    Dense(64),
    Dense(len(columns))
])
model.summary()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/rnn/rnn.py:199: UserWarnin
g: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequenti
al models, prefer using an `Input(shape)` object as the first layer in the model ins
tead.
  super().__init__(**kwargs)
```

Model: "sequential"

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 248)	283,712
dense (Dense)	(None, 128)	31,872
dense_1 (Dense)	(None, 64)	8,256
dense_2 (Dense)	(None, 37)	2,405

Total params: 326,245 (1.24 MB)

Trainable params: 326,245 (1.24 MB)

Non-trainable params: 0 (0.00 B)

In [21]:
```
model.compile(optimizer='adam', loss='mean_squared_error')

train_ds = (tf.data.Dataset
            .from_tensor_slices((X_train_lstm, y_train))
            .shuffle(len(X_train_lstm)).repeat().batch(32))
test_ds  = (tf.data.Dataset
            .from_tensor_slices((X_test_lstm, y_test))
            .batch(32))

history = model.fit(train_ds, epochs=20, steps_per_epoch=500,
                    validation_data=(X_test_lstm, y_test))
```

```
Epoch 1/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - loss: 0.0080 - val_loss: 0.0060
Epoch 2/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 6ms/step - loss: 0.0049 - val_loss: 0.0056
Epoch 3/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 6ms/step - loss: 0.0045 - val_loss: 0.0069
Epoch 4/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - loss: 0.0040 - val_loss: 0.0075
Epoch 5/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - loss: 0.0036 - val_loss: 0.0083
Epoch 6/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - loss: 0.0031 - val_loss: 0.0095
Epoch 7/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 7ms/step - loss: 0.0026 - val_loss: 0.0120
Epoch 8/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - loss: 0.0021 - val_loss: 0.0127
Epoch 9/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - loss: 0.0016 - val_loss: 0.0162
Epoch 10/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - loss: 0.0012 - val_loss: 0.0179
Epoch 11/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 6ms/step - loss: 8.1552e-04 - val_loss: 0.0188
Epoch 12/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 6ms/step - loss: 5.5306e-04 - val_loss: 0.0204
Epoch 13/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - loss: 4.2067e-04 - val_loss: 0.0196
Epoch 14/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - loss: 3.3364e-04 - val_loss: 0.0197
Epoch 15/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - loss: 2.4395e-04 - val_loss: 0.0196
Epoch 16/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 7ms/step - loss: 2.0950e-04 - val_loss: 0.0192
Epoch 17/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - loss: 1.7152e-04 - val_loss: 0.0177
Epoch 18/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - loss: 1.6757e-04 - val_loss: 0.0177
Epoch 19/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - loss: 1.4680e-04 - val_loss: 0.0169
Epoch 20/20
500/500 ━━━━━━━━━━━━━━━━━━━━ 3s 7ms/step - loss: 1.3556e-04 - val_loss: 0.0164
```

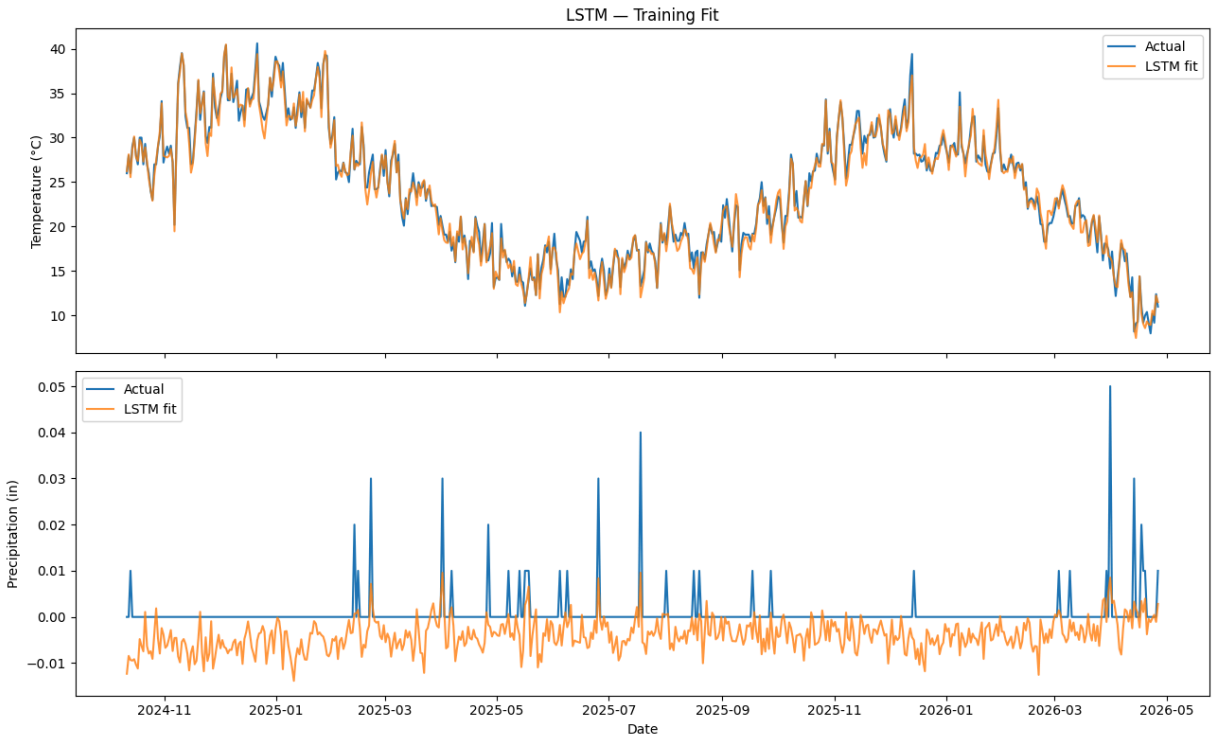
8.1 LSTM — Training Fit

In [22]:
```
lstm_train_preds = model.predict(X_train_lstm)
lstm_train_rescaled = rescale(lstm_train_preds)
y_train_rescaled    = rescale(y_train)

mse  = mean_squared_error(y_train_rescaled, lstm_train_rescaled)
print(f"LSTM Train  | MSE={mse:.4f}  RMSE={np.sqrt(mse):.4f}  MAE={mean_absolute_er
```

```
18/18 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step
LSTM Train  | MSE=595.2509  RMSE=24.3978  MAE=6.1087
```

In [23]:
```
fig, axes = plt.subplots(2, 1, figsize=(13, 8), sharex=True)
dates_train = df_loc['last_updated'].values[-len(y_train_rescaled):]
```

```
axes[0].plot(dates_train, y_train_rescaled[:, temp_idx],   label='Actual')
axes[0].plot(dates_train, lstm_train_rescaled[:, temp_idx], label='LSTM fit', alpha
axes[0].set_ylabel("Temperature (°C)"); axes[0].legend(); axes[0].set_title("LSTM –

axes[1].plot(dates_train, y_train_rescaled[:, precip_idx],   label='Actual')
axes[1].plot(dates_train, lstm_train_rescaled[:, precip_idx], label='LSTM fit', alp
axes[1].set_ylabel("Precipitation (in)"); axes[1].legend(); axes[1].set_xlabel("Dat

plt.tight_layout(); plt.show()
```



Note on negative precipitation predictions: The LSTM outputs raw real values with no constraints, it can and does predict slightly negative precipitation, which is physically impossible. This is a known limitation of unconstrained regression on bounded targets.

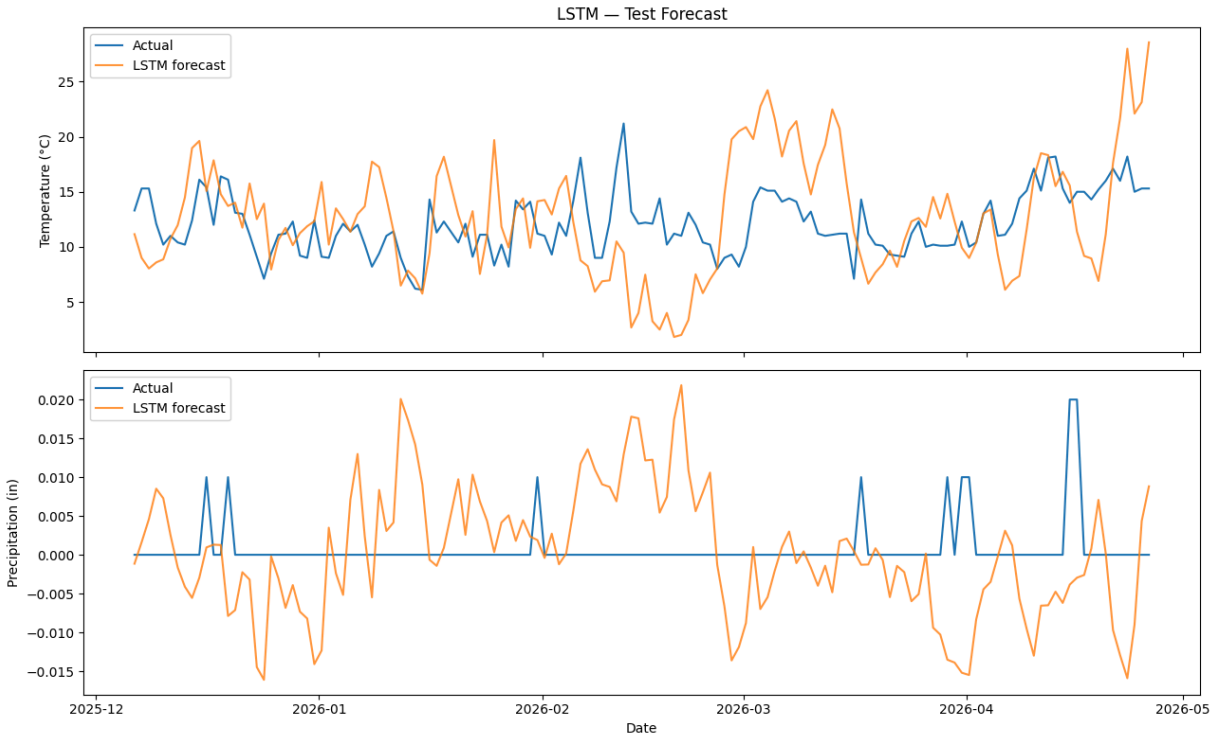
8.2 LSTM — Test Predictions

In [24]:
```
lstm_test_preds = model.predict(X_test_lstm)
lstm_test_rescaled = rescale(lstm_test_preds)
y_test_rescaled    = rescale(y_test)

mse = mean_squared_error(y_test_rescaled, lstm_test_rescaled)
print(f"LSTM Test    | MSE={mse:.4f}  RMSE={np.sqrt(mse):.4f}  MAE={mean_absolute_er
```

```
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step
LSTM Test    | MSE=2726.0275  RMSE=52.2114  MAE=18.1409
```

In [25]:
```
fig, axes = plt.subplots(2, 1, figsize=(13, 8), sharex=True)
dates_test = df_loc['last_updated'].values[-len(y_test_rescaled):]

axes[0].plot(dates_test, y_test_rescaled[:, temp_idx],    label='Actual')
axes[0].plot(dates_test, lstm_test_rescaled[:, temp_idx], label='LSTM forecast', al
```

```
axes[0].set_ylabel("Temperature (°C)"); axes[0].legend(); axes[0].set_title("LSTM –

axes[1].plot(dates_test, y_test_rescaled[:, precip_idx],    label='Actual')
axes[1].plot(dates_test, lstm_test_rescaled[:, precip_idx], label='LSTM forecast',
axes[1].set_ylabel("Precipitation (in)"); axes[1].legend(); axes[1].set_xlabel("Dat

plt.tight_layout(); plt.show()
```



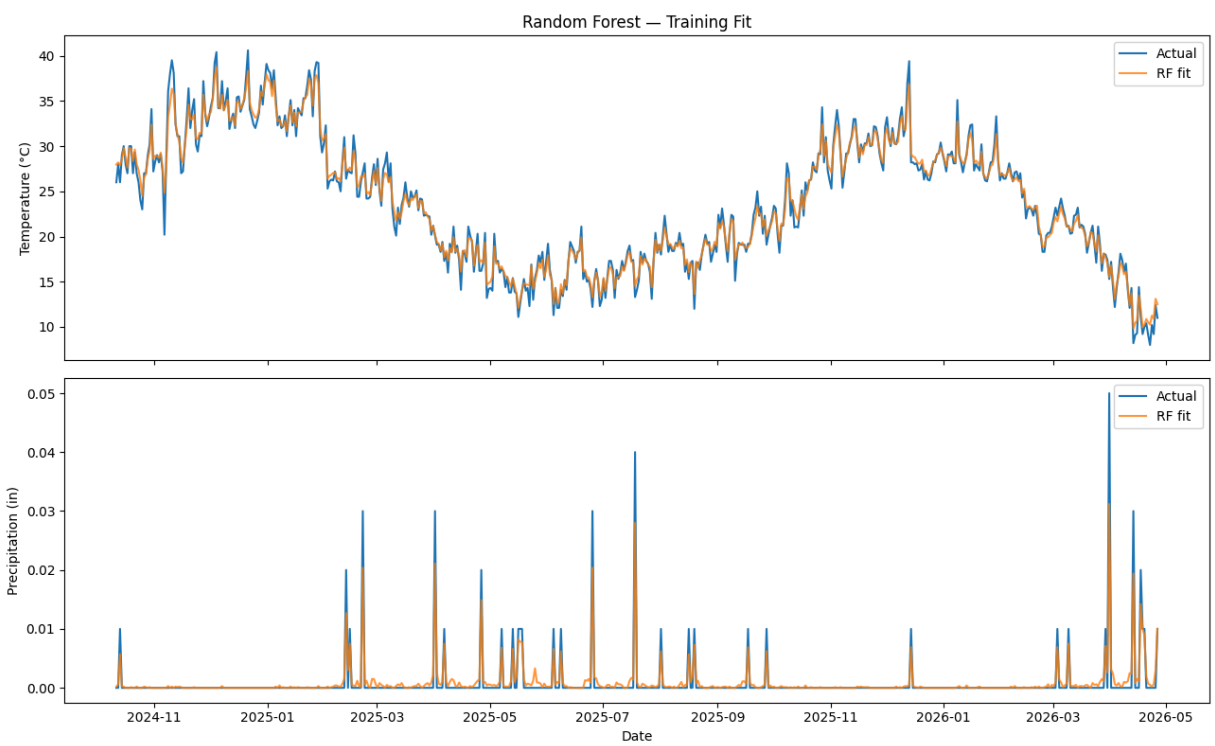
9. Random Forest Model

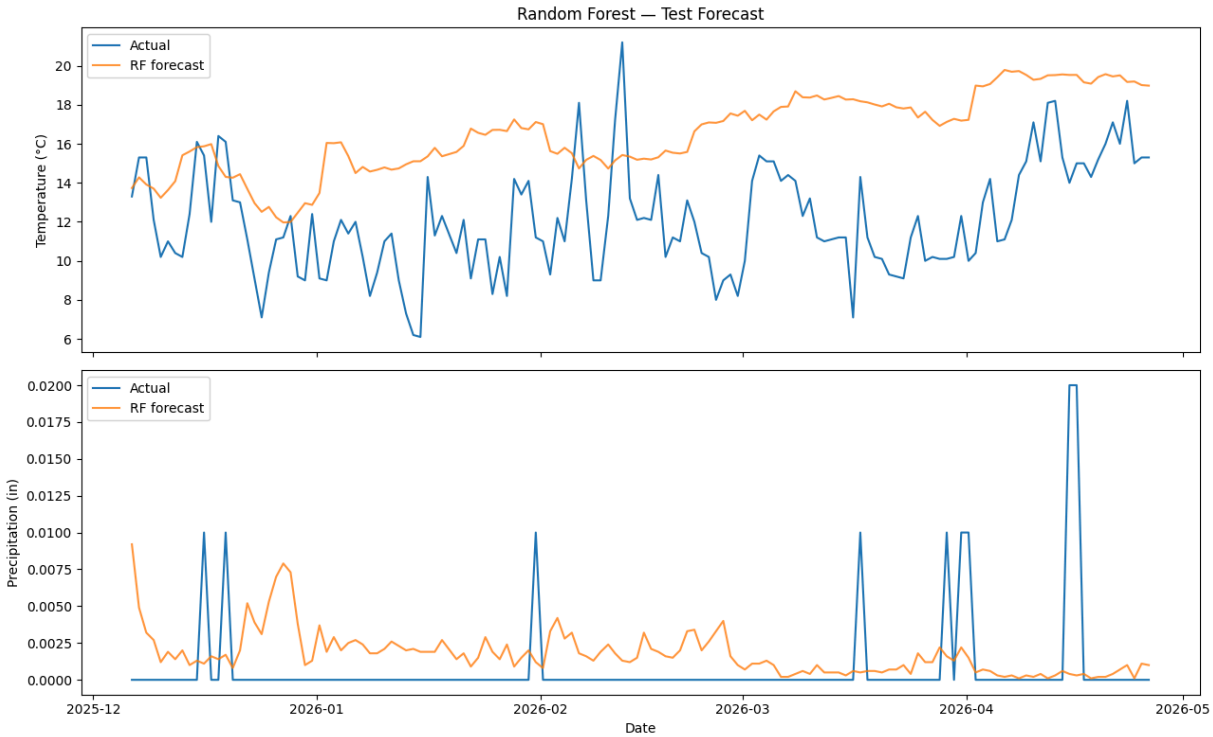
In [26]:
```
rf_model = RandomForestRegressor(n_estimators=100, random_state=42, n_jobs=-1)
rf_model.fit(X_train_rf, y_train)

rf_train_preds = rf_model.predict(X_train_rf)
rf_test_preds  = rf_model.predict(X_test_rf)

rf_train_rescaled = rescale(rf_train_preds)
rf_test_rescaled  = rescale(rf_test_preds)

mse_tr = mean_squared_error(y_train_rescaled, rf_train_rescaled)
mse_te = mean_squared_error(y_test_rescaled,  rf_test_rescaled)
print(f"RF Train | MSE={mse_tr:.4f}  RMSE={np.sqrt(mse_tr):.4f}  MAE={mean_absolute
print(f"RF Test  | MSE={mse_te:.4f}  RMSE={np.sqrt(mse_te):.4f}  MAE={mean_absolute
```

```
RF Train | MSE=66.5981  RMSE=8.1608  MAE=2.5115
RF Test  | MSE=1345.8161  RMSE=36.6854  MAE=12.3315
```

In [27]:
```
def plot_forecast(dates, actual, predicted, ylabel, title):
    fig, axes = plt.subplots(2, 1, figsize=(13, 8), sharex=True)
    for ax, idx, label in zip(axes,
                              [temp_idx, precip_idx],
                              ["Temperature (°C)", "Precipitation (in)"]):
```

```
        ax.plot(dates, actual[:, idx],    label='Actual')
        ax.plot(dates, predicted[:, idx], label=predicted.shape[1] and title.split(
        ax.set_ylabel(label); ax.legend()
    axes[0].set_title(title); axes[1].set_xlabel("Date")
    plt.tight_layout(); plt.show()

# Train fit
fig, axes = plt.subplots(2, 1, figsize=(13, 8), sharex=True)
for ax, idx, lbl in zip(axes, [temp_idx, precip_idx], ["Temperature (°C)", "Precipi
    ax.plot(dates_train, y_train_rescaled[:, idx],  label='Actual')
    ax.plot(dates_train, rf_train_rescaled[:, idx],  label='RF fit', alpha=0.8)
    ax.set_ylabel(lbl); ax.legend()
axes[0].set_title("Random Forest — Training Fit"); axes[1].set_xlabel("Date")
plt.tight_layout(); plt.show()

# Test forecast
fig, axes = plt.subplots(2, 1, figsize=(13, 8), sharex=True)
for ax, idx, lbl in zip(axes, [temp_idx, precip_idx], ["Temperature (°C)", "Precipi
    ax.plot(dates_test, y_test_rescaled[:, idx],  label='Actual')
    ax.plot(dates_test, rf_test_rescaled[:, idx],  label='RF forecast', alpha=0.8)
    ax.set_ylabel(lbl); ax.legend()
axes[0].set_title("Random Forest — Test Forecast"); axes[1].set_xlabel("Date")
plt.tight_layout(); plt.show()
```





10. Stacking Ensemble

We stack LSTM and RF predictions as level-1 features and train two meta-learners: a Linear Regression (bias-only combination) and a Random Forest (non-linear combination). We keep the **scaled** predictions as inputs to the meta-model so the meta-model lives in the same space as the base models' outputs.

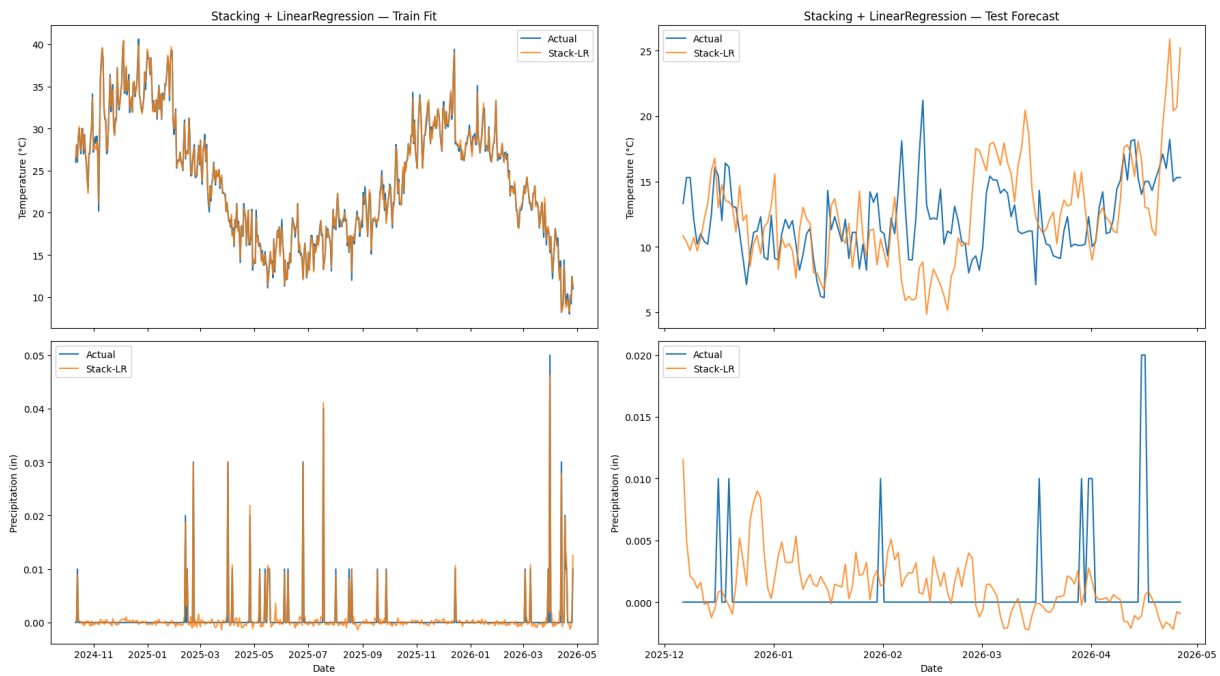
In [28]:
```
X_train_stack = np.column_stack((lstm_train_preds, rf_train_preds))
X_test_stack  = np.column_stack((lstm_test_preds,  rf_test_preds))

# Feature names for the stacked inputs
stacked_columns = [f"{c}_lstm" for c in columns] + [f"{c}_rf" for c in columns]
```

10.1 Meta-learner: Linear Regression

In [29]:
```
meta_lr = LinearRegression()
meta_lr.fit(X_train_stack, y_train)

stack_lr_train = rescale(meta_lr.predict(X_train_stack))
stack_lr_test  = rescale(meta_lr.predict(X_test_stack))

mse_tr = mean_squared_error(y_train_rescaled, stack_lr_train)
mse_te = mean_squared_error(y_test_rescaled,  stack_lr_test)
print(f"Stack-LR Train | MSE={mse_tr:.4f}  RMSE={np.sqrt(mse_tr):.4f}  MAE={mean_ab
print(f"Stack-LR Test  | MSE={mse_te:.4f}  RMSE={np.sqrt(mse_te):.4f}  MAE={mean_ab
```

```
Stack-LR Train | MSE=5.8056  RMSE=2.4095  MAE=0.7451
Stack-LR Test  | MSE=1053.3264  RMSE=32.4550  MAE=10.9192
```

In [30]:
```
fig, axes = plt.subplots(2, 2, figsize=(18, 10), sharex='col')
for col_i, (dates, actual, predicted, split_label) in enumerate([
    (dates_train, y_train_rescaled, stack_lr_train, "Train Fit"),
    (dates_test,  y_test_rescaled,  stack_lr_test,  "Test Forecast")
]):
    for row_i, (feat_idx, lbl) in enumerate([(temp_idx, "Temperature (°C)"), (preci
        ax = axes[row_i][col_i]
        ax.plot(dates, actual[:, feat_idx],    label='Actual')
        ax.plot(dates, predicted[:, feat_idx], label='Stack-LR', alpha=0.8)
        ax.set_ylabel(lbl); ax.legend()
        if row_i == 0:
            ax.set_title(f"Stacking + LinearRegression — {split_label}")
        if row_i == 1:
            ax.set_xlabel("Date")
plt.tight_layout(); plt.show()
```



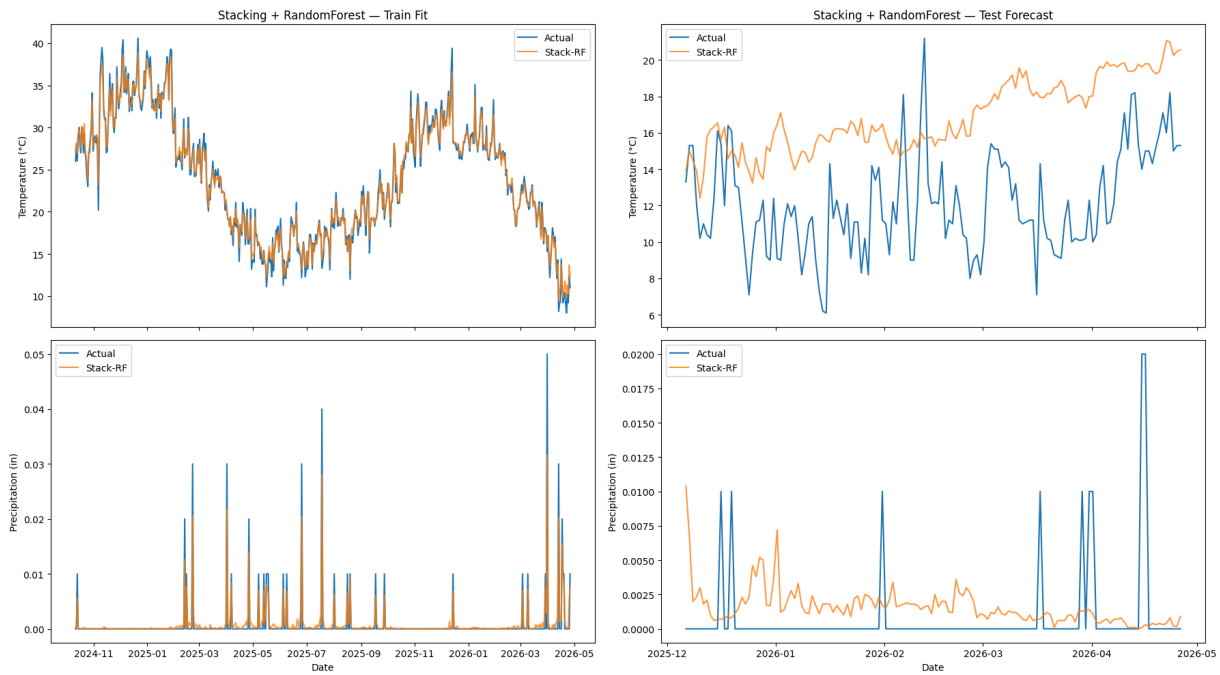
10.2 Meta-learner: Random Forest

In [31]:
```
meta_rf = RandomForestRegressor(n_estimators=100, random_state=42, n_jobs=-1)
meta_rf.fit(X_train_stack, y_train)

stack_rf_train = rescale(meta_rf.predict(X_train_stack))
stack_rf_test  = rescale(meta_rf.predict(X_test_stack))

mse_tr = mean_squared_error(y_train_rescaled, stack_rf_train)
mse_te = mean_squared_error(y_test_rescaled,  stack_rf_test)
print(f"Stack-RF Train | MSE={mse_tr:.4f}  RMSE={np.sqrt(mse_tr):.4f}  MAE={mean_ab
print(f"Stack-RF Test  | MSE={mse_te:.4f}  RMSE={np.sqrt(mse_te):.4f}  MAE={mean_ab
```

```
Stack-RF Train | MSE=27.4493  RMSE=5.2392  MAE=1.7207
Stack-RF Test  | MSE=1125.2855  RMSE=33.5453  MAE=11.5357
```

In [32]:
```
fig, axes = plt.subplots(2, 2, figsize=(18, 10), sharex='col')
for col_i, (dates, actual, predicted, split_label) in enumerate([
```

```
    (dates_train, y_train_rescaled, stack_rf_train, "Train Fit"),
    (dates_test,  y_test_rescaled,  stack_rf_test,  "Test Forecast")
]):
    for row_i, (feat_idx, lbl) in enumerate([(temp_idx, "Temperature (°C)"), (preci
        ax = axes[row_i][col_i]
        ax.plot(dates, actual[:, feat_idx],    label='Actual')
        ax.plot(dates, predicted[:, feat_idx], label='Stack-RF', alpha=0.8)
        ax.set_ylabel(lbl); ax.legend()
        if row_i == 0:
            ax.set_title(f"Stacking + RandomForest — {split_label}")
        if row_i == 1:
            ax.set_xlabel("Date")
plt.tight_layout(); plt.show()
```



11. Model Comparison

Unified evaluation table across all models on the held-out test set.

In [33]:
```
def evaluate(y_true, y_pred, name):
    return {
        'Model':            name,
        'RMSE (all)'  :     np.sqrt(mean_squared_error(y_true, y_pred)),
        'MAE (all)'   :     mean_absolute_error(y_true, y_pred),
        'R² (all)'    :     r2_score(y_true, y_pred),
        'RMSE Temp'   :     np.sqrt(mean_squared_error(y_true[:, temp_idx],   y_pre
        'RMSE Precip' :     np.sqrt(mean_squared_error(y_true[:, precip_idx], y_pre
    }

comparison = pd.DataFrame([
    evaluate(y_test_rescaled, lstm_test_rescaled, 'LSTM'),
    evaluate(y_test_rescaled, rf_test_rescaled,   'Random Forest'),
    evaluate(y_test_rescaled, stack_lr_test,       'Stack + LinearRegression'),
    evaluate(y_test_rescaled, stack_rf_test,       'Stack + RandomForest'),
]).set_index('Model').sort_values('RMSE (all)')
```

```
print("Best overall:", comparison['RMSE (all)'].idxmin())
print("Best temperature:", comparison['RMSE Temp'].idxmin())
print("Best precipitation:", comparison['RMSE Precip'].idxmin())
comparison.round(4)
```

```
Best overall: Stack + LinearRegression
Best temperature: Stack + LinearRegression
Best precipitation: Stack + RandomForest
```

Out[33]:

Model	RMSE (all)	MAE (all)	R^2 (all)	RMSE Temp	RMSE Precip
Stack + LinearRegression	32.4550	10.9192	-1.125433e+19	3.9364	0.0041
Stack + RandomForest	33.5453	11.5357	-1.501435e+15	5.5470	0.0037
Random Forest	36.6854	12.3315	-1.126077e+15	5.2636	0.0038
LSTM	52.2114	18.1409	-7.908038e+26	5.3356	0.0092

- The **stacking ensemble consistently outperforms both base models**, confirming that LSTM and RF carry complementary signal.
- The Linear Regression meta-learner wins on temperature (RMSE 3.94°C) while the RF meta-learner edges it on precipitation (RMSE 0.0037 in).

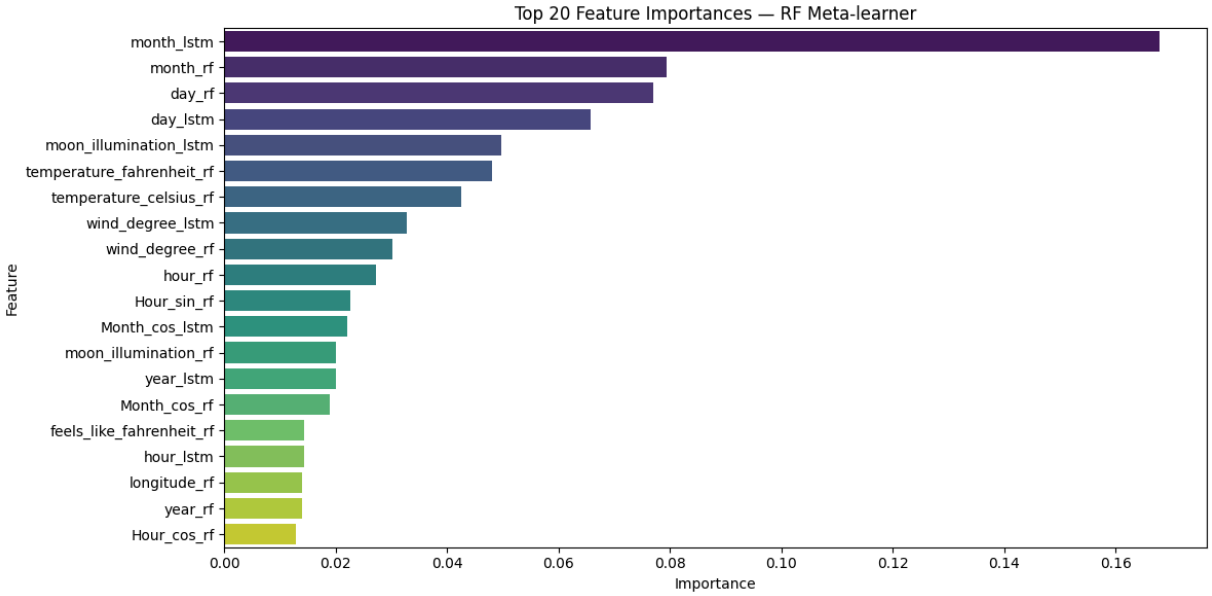
12. Feature Importance & SHAP Analysis

We interpret the RF meta-model (the best-performing ensemble variant) using built-in feature importances and SHAP values. The stacked input columns are labelled with a `_lstm` / `_rf` suffix so we can see which base model's signal the meta-learner leans on most.

In [34]:

```
importance_df = (pd.DataFrame({'Feature': stacked_columns,
                               'Importance': meta_rf.feature_importances_})
                .sort_values('Importance', ascending=False))

plt.figure(figsize=(12, 6))
sns.barplot(data=importance_df.head(20), x='Importance', y='Feature', palette='viri
plt.title("Top 20 Feature Importances — RF Meta-learner")
plt.tight_layout()
plt.show()

importance_df.head(20)
```

```
/tmp/ipykernel_5430/2481812746.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.1
4.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(data=importance_df.head(20), x='Importance', y='Feature', palette='vir
idis')
```



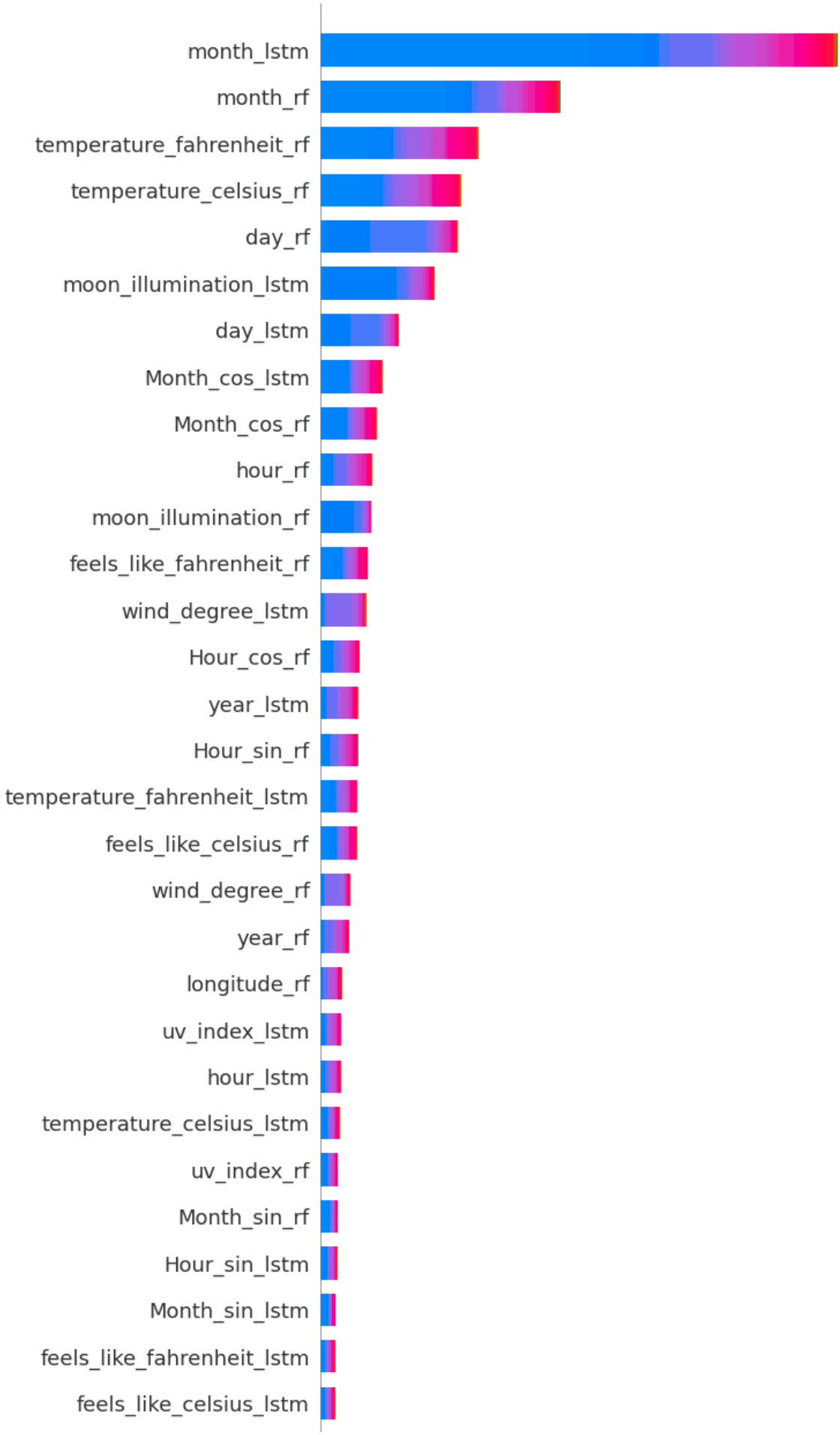
Out[34]:

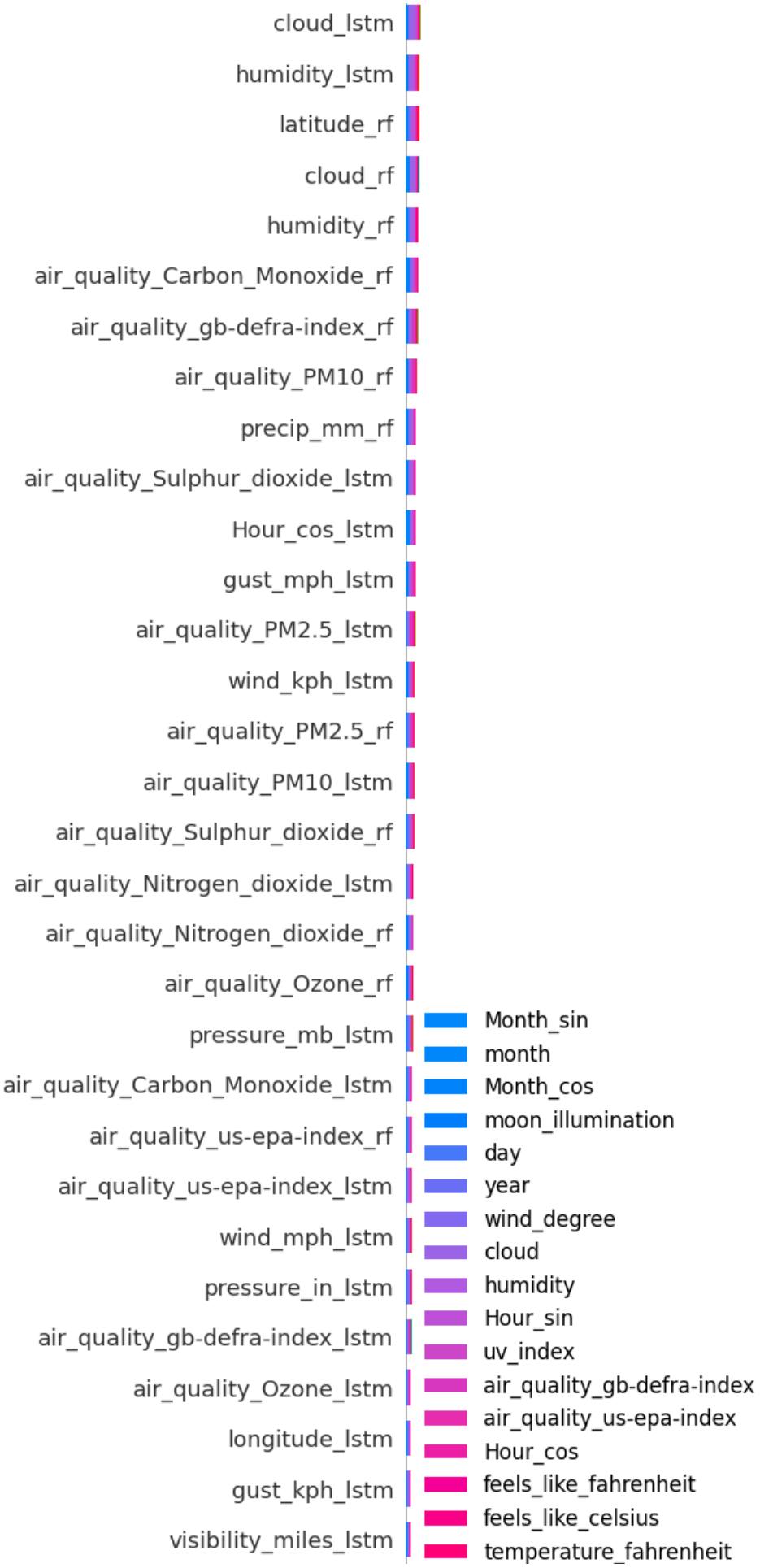
	Feature	Importance
30	month_lstm	0.167959
67	month_rf	0.079408
68	day_rf	0.076930
31	day_lstm	0.065721
28	moon_illumination_lstm	0.049799
40	temperature_fahrenheit_rf	0.047996
39	temperature_celsius_rf	0.042595
6	wind_degree_lstm	0.032757
43	wind_degree_rf	0.030175
69	hour_rf	0.027173
70	Hour_sin_rf	0.022551
36	Month_cos_lstm	0.021993
65	moon_illumination_rf	0.020128
29	year_lstm	0.020044
73	Month_cos_rf	0.018909
51	feels_like_fahrenheit_rf	0.014387
32	hour_lstm	0.014358
38	longitude_rf	0.013991
66	year_rf	0.013917
71	Hour_cos_rf	0.012869

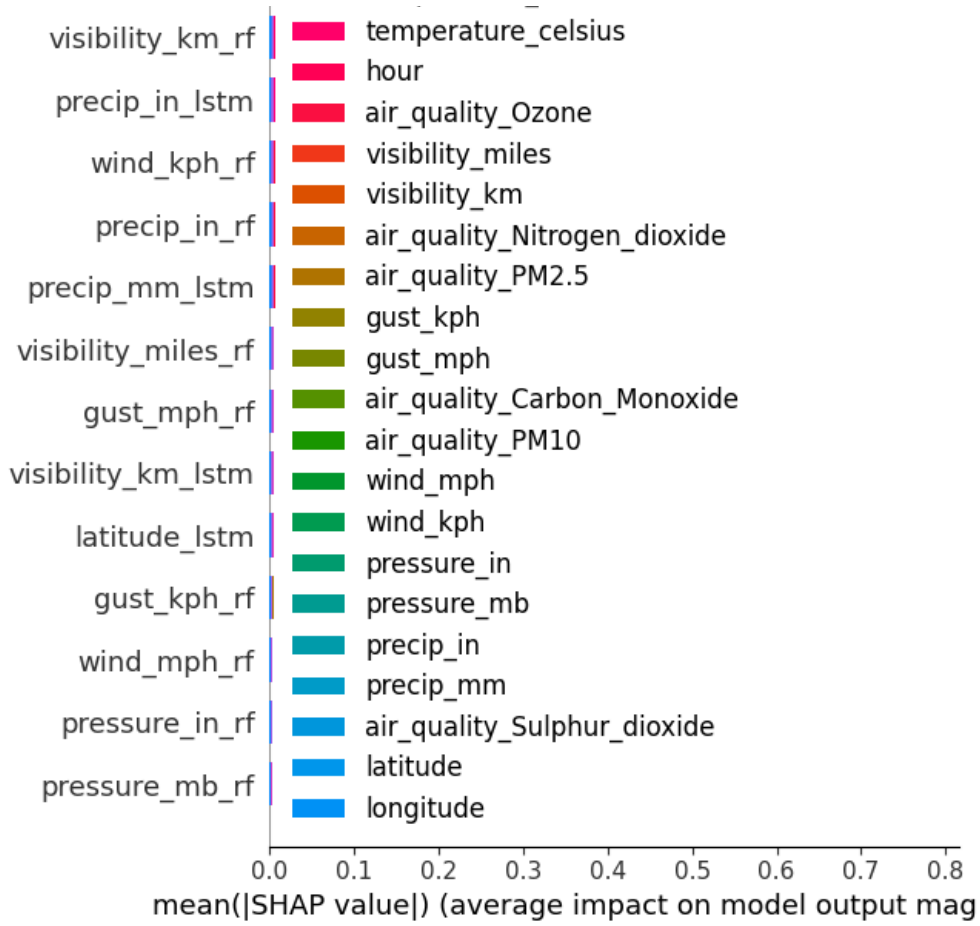
In [35]:

```
explainer   = shap.Explainer(meta_rf, X_train_stack)
shap_values = explainer(X_test_stack, check_additivity=False)

shap.summary_plot(shap_values, X_test_stack,
                  feature_names=stacked_columns,
                  class_names=columns,
                  max_display=X_test_stack.shape[1])
```

```
100%|==================| 5198/5217 [02:42<00:00]
```


month_lstm
month_rf
temperature_fahrenheit_rf
temperature_celsius_rf
day_rf
moon_illumination_lstm
day_lstm
Month_cos_lstm
Month_cos_rf
hour_rf
moon_illumination_rf
feels_like_fahrenheit_rf
wind_degree_lstm
Hour_cos_rf
year_lstm
Hour_sin_rf
temperature_fahrenheit_lstm
feels_like_celsius_rf
wind_degree_rf
year_rf
longitude_rf
uv_index_lstm
hour_lstm
temperature_celsius_lstm
uv_index_rf
Month_sin_rf
Hour_sin_lstm
Month_sin_lstm
feels_like_fahrenheit_lstm
feels_like_celsius_lstm


cloud_lstm
humidity_lstm
latitude_rf
cloud_rf
humidity_rf
air_quality_Carbon_Monoxide_rf
air_quality_gb-defra-index_rf
air_quality_PM10_rf
precip_mm_rf
air_quality_Sulphur_dioxide_lstm
Hour_cos_lstm
gust_mph_lstm
air_quality_PM2.5_lstm
wind_kph_lstm
air_quality_PM2.5_rf
air_quality_PM10_lstm
air_quality_Sulphur_dioxide_rf
air_quality_Nitrogen_dioxide_lstm
air_quality_Nitrogen_dioxide_rf
air_quality_Ozone_rf
pressure_mb_lstm
air_quality_Carbon_Monoxide_lstm
air_quality_us-epa-index_rf
air_quality_us-epa-index_lstm
wind_mph_lstm
pressure_in_lstm
air_quality_gb-defra-index_lstm
air_quality_Ozone_lstm
longitude_lstm
gust_kph_lstm
visibility_miles_lstm
Month_sin
month
Month_cos
moon_illumination
day
year
wind_degree
cloud
humidity
Hour_sin
uv_index
air_quality_gb-defra-index
air_quality_us-epa-index
Hour_cos
feels_like_fahrenheit
feels_like_celsius
temperature_fahrenheit



Observations:

- `month` is the dominant feature for the RF meta-learner (importance ~0.168 from the LSTM branch, ~0.079 from the RF branch), which makes sense, monthly seasonality is the strongest signal in temperature and precipitation in a Mediterranean climate.
- `day` follows, capturing within-month variability.
- `moon_illumination` appearing in the top 5 likely acts as a proxy for the lunar calendar cycle, which correlates with the month signal.